MAX NICOSIA, PhD

Software R&D & Systems Engineer

Tokyo, Japan | lmnicosia@gmail.com | 070 8973 9527 | lumyx2.github.io | linkedin.com/in/maxnicosia

PROFILE

Software R&D and Systems Engineer with 10+ years of hands-on experience designing, prototyping, implementing and validating complex technical systems across simulation, XR, safety-critical software, distributed systems, hardware/software integration and low-latency applications. Strongest at taking early-stage concepts from requirements and architecture through rapid prototyping, implementation, experimentation, optimisation and verification. Background includes technical leadership at Woven by Toyota, commercial R&D for the UK Ministry of Defence, high-performance C++ financial systems and a Cambridge PhD in attention-aware distributed systems.

TECHNICAL EXPERTISE

Languages	C/C++, C#, Python, Java, TypeScript, JavaScript, Ruby, R, SQL/MySQL
Systems & Networking	Distributed systems, middleware, multithreading, lockless programming, synchronisation, memory barriers, TCP/UDP/IP, WebSockets, hardware drivers, sensor integration, custom APIs, kernel-level networking
R&D & Engineering	Systems architecture, rapid prototyping, simulation, XR/VR/AR, HCI, human-in-the-loop systems, requirements elicitation, experimental design, verification & validation, test-driven development
Performance & HPC	Profiling, low-latency systems, memory management, memory pools, cache-aware design, OpenMP, MPI, pthreads, Boost
Simulation & XR	Unity, Unreal, Vehicle Physics Pro, pathfinding, splines, behaviour trees/blackboards, 2D/3D adaptive interfaces, real-time visualisation, shaders/filters, LOD, dynamic asset loading, object serialisation
Data & ML	Statistical testing/model fitting, sensor fusion, scikit-learn, PyTorch, NumPy, SciPy, Pandas, Jupyter, d3.js, MapReduce/Hadoop
Platforms & Tooling	Linux, AWS, Docker, Podman, CMake, Make, Git/GitHub Actions, Bash, PowerShell, Android/Gradle, iOS/Xcode
Leadership	Technical leadership, stakeholder management, milestone planning, project delivery, proposals/contract bids, live demonstrations and presentations

PROFESSIONAL EXPERIENCE

Woven by Toyota — **Jun 2024 – Mar 2026**

Technical Lead (XR Safety / Driver Sync Assist) — Tokyo, Japan

- Technical lead for the judgement and simulation workstream of an XR / driver-safety R&D programme, leading up to 2 engineers and coordinating objectives, task delegation, blockers and milestone delivery.
- Re-architected an inherited Unity environment into a structured C# simulation platform, introducing interfaces, inheritance, dependency injection and the Humble Object pattern to improve maintainability and enable automated testing.
- Developed automated tests covering vehicle-physics calculations and safety judgement for vehicles, pedestrians, traffic lights and other obstacles; reviewed pull requests and supported stable development/deployment environments.
- Designed safety judgement using live and simulated data, including probabilistic collision risk, driver reaction time and cognitive-load considerations; developed reproducible AI vehicle/pedestrian pathfinding scenarios.
- Integrated Vehicle Physics Pro for realistic vehicle/engine physics and implemented scenario capture, playback and re-simulation of scenarios and real-world collected data.
- Developed C/C++ wrappers and CAN integration for steering, speed, acceleration, pedals, feature buttons and other vehicle/simulation controls and sensor interfaces.
- Profiled and optimised simulation performance for a 60+ FPS target, typically reaching approximately 90–120 FPS depending on scene complexity while supporting 5 x 4K displays plus 2 x HD mirrored displays.
- Designed and executed V&V experiments with 16 participants using Latin-square counterbalancing; defined testing and product-verification strategies supporting progression through PoC and beta gates into early production development.
- Presented quarterly technical demonstrations to product management and co-invented visualisation technology resulting in Japanese and US patent filings.

Cambridge Intelligent Systems UK Ltd. **Nov 2017 – Jun 2025**

CEO, Co-founder & Technical Lead
Cambridge, UK

- Co-founded an R&D company delivering applied research, simulation systems and prototypes for the UK Ministry of Defence; served as technical lead and principal software developer across all projects.
- Secured approximately £150k in combined funding across two MoD R&D programmes involving XR, simulation, attention-aware systems and human-machine interaction.
- Took projects from proposal and requirements through architecture, implementation, hardware/software integration, trials, stakeholder demonstration and delivery.
- Designed and developed the complete software systems for both MoD projects in Unity/C#, working directly with DSTL/MoD engineers and researchers throughout requirements gathering, development and evaluation.
- Personally ran technical demonstrations and trials with MoD/DSTL and military stakeholders; managed milestones, deliverables, procurement, stakeholder communication, proposals and contract bids.
- Delivered mixed-reality armoured-vehicle controls and an attention-aware VR threat-detection/training system. See Selected R&D Projects for technical details.

FinstadiumX **Apr 2024 – May 2024**

Strategic Development Team Lead
Tokyo, Japan

- Led a 2-person development team working on production low-latency C++ systems for financial trading applications.
- Designed, developed and deployed production C++ components achieving approximately 20 ms latency while processing 2,000–5,000 transactions per second; maintained services supporting products handling approximately JPY 7–10B in daily revenue per product.
- Applied low-latency techniques including memory barriers, ring buffers, lockless programming, memory pools, cache alignment and compiler optimisation; profiled legacy components for redesign and performance improvement.
- Developed automated CMake-based build, test and deployment tooling for Linux and Windows, improving deployment reliability and preventing critical integration issues.
- Worked with FIX-based financial systems and Docker/Podman environments; reorganised development work around engineers' technical strengths and began formalising latency/performance testing.

Tokyo Academics **Nov 2022 – Mar 2024**

Head of Research
Tokyo, Japan

- Led the research division, managing approximately 20 part-time researchers and overseeing research delivery, operations, sales and marketing.
- Grew the research programme from approximately 20 to around 50–60 students and achieved approximately 125% growth in sales receipts against a 50% growth target.
- Recruited and supervised researchers across technical and academic projects; managed project allocation, client requirements, researcher performance and delivery quality.
- Supervised computer-science research and development projects including software prototyping and machine-learning applications.

University of Cambridge **Oct 2019 – Jul 2020**

Research Associate – Department of Engineering
Cambridge, UK

- Supported research in attention-aware and human-computer interaction systems alongside completion and experimental validation of PhD research.
- Maintained research databases and developed Bash scripts and supporting tooling for data management and research workflows.
- Assisted with research infrastructure, software development, technical support and experimental systems across the group.

Cambridge University Technical Services **May 2017 – Dec 2017**

Software R&D Consultant
Cambridge, UK

- Adapted technology originating from my PhD into a UK MoD-funded C++ distributed attention-monitoring platform for complex multi-display environments, with air traffic control as the primary use case.
- Designed middleware for sensor acquisition, aggregation and conversion into global attention events; integrated Tobii eye trackers and exposed functionality through a purpose-built API.
- Deployed the system across 5 application computers with 5 eye trackers plus coordination and time-server infrastructure; developed JavaScript/TypeScript and d3.js live visualisation software.
- Worked with DSTL, MoD and NATS stakeholders on requirements and technical demonstrations. See ACC101965 under Selected R&D Projects.

University of St Andrews **Jan 2012 – Sep 2012**

Research Assistant – Computer Science
St Andrews, UK

- Conducted HCI research investigating Fitts' Law and human perception on large interactive displays; developed C#/XNA experiments for Microsoft Surface PixelSense 2.0.
- Analysed experimental data and tested statistical models in R and Python/matplotlib. Research contributed to the ACM CHI 2014 publication listed below.

University of St Andrews **Jun 2011 – Sep 2011**

iGEM Research Intern St Andrews, UK

- Funded member of the university's synthetic-biology team; contributed to a biological kill-switch/drug-delivery concept, MATLAB reaction-rate modelling, laboratory work and technical documentation. Team received an international iGEM Gold Medal.

University of St Andrews **May 2011 – Jun 2011**

Research Assistant St Andrews, UK

- Developed a Python simulator modelling mosquito disease transmission, population dynamics and resistance, including constrained graphs for mating patterns; automated R analysis/plots with AWK and Bash.
- Contributed to a separate animal-tracking project using clustering algorithms to identify frequented locations and behavioural patterns from seal movement data.

University of St Andrews **Aug 2010 – Sep 2010**

Software Developer St Andrews, UK

- Developed OpenMath binary-encoding functionality for the GAP computer algebra system.

Technical University of Denmark **Jun 2010 – Aug 2010**

C Developer Lyngby, Denmark

- Developed accessibility functionality for the GNOME/Linux desktop, including Evince integration with the Orca screen reader, and produced supporting documentation for the Vinux accessible Linux distribution.

SELECTED R&D PROJECTS

ACC2006330 – Attention-Aware Mixed/AR Controls **Mar 2020 – Feb 2021**

UK Ministry of Defence / DSTL Cambridge Intelligent Systems

Developed an attention-aware VR simulation and training system to investigate operator responses to configurable threats and adaptive visualisation aids in armoured-vehicle environments. Principal software developer, responsible for Unity/C# implementation and direct stakeholder work from requirements through trials and demonstrations.

- Built configurable urban and open-terrain VR scenarios using Oculus Rift and UltraLeap hand interaction, with human/vehicle NPCs, pathfinding, spawning, threat escalation, animation and audio.
- Implemented parameterised threat numbers, timing, routes and visualisation behaviour, together with NATO symbology and threat tagging across 3D and map-based interfaces.
- Developed gaze-driven attention aids including attended/unattended-area tracking, out-of-view threat pointers and animated alerts for unattended threats.
- Delivered interim and final live demonstrations to DSTL, MoD and armed-forces stakeholders. Technology from the project was subsequently incorporated into MoD tank-operation training practices.

ACC2000981 – Mixed Reality Controls for Armoured Vehicles **Dec 2018 – Oct 2019**

UK Ministry of Defence / DSTL Cambridge Intelligent Systems

Developed a mixed-reality vehicle-interface prototype investigating replacement or augmentation of physical controls in armoured vehicles with virtual interfaces. VR was used to simulate the intended AR deployment.

- Developed GVA-compliant interactive virtual interfaces and menus in Unity/C#, using Oculus Rift and UltraLeap hand tracking.
- Implemented passthrough functionality for operator arms/hands, simulated 360-degree internal/external cameras with gesture manipulation, multiple viewpoints and camera axes.
- Built a head-up interface integrating vehicle information, compass, minimap, threats, virtual controls, NATO threat markers and pointers.
- Delivered live demonstrations to DSTL, MoD and armed-forces stakeholders; concepts were subsequently incorporated into future vehicle-development work.

ACC101965 – Distributed Attention Monitoring **May 2017 – Dec 2017**

UK Ministry of Defence / DSTL Cambridge University Technical Services

Adapted technology originating from my PhD into a distributed attention-monitoring system for complex multi-display environments, with air traffic control as the primary use case.

- Developed C/C++ middleware receiving eye-tracking data from multiple networked computers and aggregating it into global attention events.
- Deployed across 5 application computers with 5 eye trackers plus dedicated coordination and time-server infrastructure; integrated Tobii drivers, data fusion and configurable attention discretisation.
- Built a purpose-designed API for application subscriptions/hooks and JavaScript/d3.js overlays for unattended information, data age and off-screen changes.
- Delivered six software packages, technical reports and a live stakeholder demonstration; worked with DSTL, MoD and NATS on requirements and technical briefings.

PHD RESEARCH & RESEARCH ENGINEERING

Attention Management System **2014 – 2023**

University of Cambridge Department of Engineering

Designed, implemented and evaluated a distributed attention-management architecture for complex multi-display, human-in-the-loop environments. The system combined sensor data, application state and live performance information to identify attentional state and trigger adaptive interventions.

- Designed a multi-component middleware architecture spanning networked computers, structured-light sensors and eye trackers, with application integration through purpose-built interfaces.
- Implemented multithreaded C++ sensor networking and aggregation components interacting directly with structured-light sensors and eye-tracker drivers.
- Developed Python monitoring and simulation components using TCP and WebSockets to connect sensor information with TypeScript web applications.
- Developed TypeScript/React and d3.js tools for live and retrospective visualisation of operator performance and attention.
- Built a Python simulator capable of replaying captured operator actions to model alternative operational outcomes under different visualisation/intervention policies.
- Evaluated the integrated system through human-participant experiments and simulation using collected task-performance data; results were published in peer-reviewed venues.

Additional Research Engineering

- Contributed reusable research-group tooling including d3.js live graphing, Unity prototyping interfaces, networking libraries, sensor polling and machine-learning model development.
- Deployed and maintained VMs and Docker containers on local research servers and developed Python/psutil monitoring scripts for resource usage and uptime.
- Used AWS resources for sensor-data machine-learning experiments and parameter-search workflows.
- Developed an API exposing operator head pose in a 5 m x 5 m room instrumented with 8 Microsoft Kinect 2.0 sensors using a purpose-built machine-learning classifier; work contributed to HCCS 2018.
- Developed Android applications for collecting data from Fitts' Law experiments used in research-group publications.

EDUCATION

PhD in Engineering **2014 – 2023**

University of Cambridge Cambridge, UK

Thesis: Design, Implementation and Evaluation of an Attention Management System.

- Designed, implemented and experimentally validated a multi-component attention-management architecture for complex multi-display, human-in-the-loop environments.
- Architected a distributed system combining networked computers, eye trackers and structured-light sensors to detect operator attentional state, associate attention with application data changes and trigger adaptive interventions.
- Developed multithreaded C++ sensor/data-aggregation middleware, Python attention-management and simulation software, and TypeScript/JavaScript experimental applications and visualisation tools.
- Designed and conducted controlled human-participant experiments and simulation studies. Results showed significant improvements in primary and secondary task performance in 3 of 4 trials.
- Research subsequently contributed to UK Ministry of Defence R&D projects and peer-reviewed publications.

MPhil in Advanced Computer Science **2012 – 2013**

University of Cambridge Cambridge, UK

Thesis: A privacy-preserving advertisement delivery system. Developed a Python server and Chrome extension implementing a privacy-preserving advertising architecture using publish/subscribe communication, WebSockets and RSA. Coursework included Security, Data-Centric Networking, Network Theory, Distributed Networks and Mobile Application Development.

BSc in Computer Science **2008 – 2012**

University of St Andrews St Andrews, UK

Dissertation: developed a reduced implementation of the INLPv4 protocol (RFC 6740) in C as a Linux kernel-level module. Studies included systems programming, software engineering, networking and computer-science research.

PUBLICATIONS & PATENTS

- Nicosia, M., & Kristensson, P. O. (2024). "Risk management in human-in-the-loop AI-assisted attention aware systems." In *Putting AI in the Critical Loop*, pp. 81–92. Academic Press.
- Nicosia, M. and Kristensson, P. O. (2021). "Design principles for AI-assisted attention aware systems in human-in-the-loop safety critical applications." In *Engineering Artificially Intelligent Systems*. Springer Nature.
- Nicosia, M. and Kristensson, P. O. (2020). "A conceptual design of an inattention management middleware with adaptive target saliency." *IEEE Aerospace Conference*.
- Nicosia, M. and Kristensson, P. O. (2018). "Inattention management middleware for human-in-the-loop multi-display applications." *IEEE HCCS 2018*, 71–76.
- Nicosia, M., Oulasvirta, A. and Kristensson, P. O. (2014). "Modeling the perception of user performance." *ACM CHI 2014*, 1747–1756.

Patents

- Nicosia, L. M., & Fukui, Y. (April 2025). *USER STATUS NOTIFICATION METHOD AND USER SUPPORT SYSTEM*. Japanese Application No. 2025-063631.
- Nicosia, L. M., & Fukui, Y. (January 2026). *USER STATUS NOTIFICATION METHOD AND USER SUPPORT SYSTEM*. US Application filed.

ADDITIONAL PROJECT

Adaptive Learning Prototype

Developed a prototype iOS mobile application in Unreal using machine learning / Bayesian optimisation to adapt learning activities based on user performance within a problem space.

LANGUAGES

English (Native) | Spanish (Fluent) | German (Business) | Japanese (Very basic)